OC Core 1.3.2 Methods and Reproducibility Companion

Release-Governed Evidence, Replay, and Audit Route

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Scope

This companion describes how OC Core 1.3.2 connects scientific claims to reproducible release artifacts. It is a release-bound methods surface, not a separate theory authority.

Methodological Rule

Every release-visible scientific assertion must be mapped to one of the following support routes:

- proof or source-audited theorem support;
- replay pass with fixed input, code route, output hash, and tolerance envelope;
- bounded operational support with explicit falsifier and claim ceiling;
- strict salvage state when a line is historically relevant but not Core authority;
- owner-gated publication or value externalization state when the blocker is not a scientific theorem.

Reproducibility Surfaces

Surface	Reproducibility role
Release machine package/evaluate	Rebuilds package metadata, manifests, checksums, ZIP integrity report, and no-send verdict.
PDF quality gate	Extracts text from primary PDFs and fails on unresolved references, local paths, stale version strings, stale closure language, missing release identity, or missing master dedication.
K0–K12 projection state	Records whether each domain projection is replay-backed, source-audited, or bounded as structural/operator signature.
DRT strict salvage	Keeps DRT outside Core authority unless workflow, figures, PDF, hashes, and Core-gate reconciliation pass.
Tsukrov repair	Preserves private reviewer-source material outside the package while exposing public-safe response routes and release references.

Parfitian Cerberus	Blocks release readiness when ethical/release-governance findings exceed the configured threshold.
LRGEF	Controls publication locks, owner approval, provenance, source freshness, and release artifact inventory.

Prediction and Quantitative Language

The word "prediction" is release-permitted only when the corresponding artifact includes protocol, code, data or fixture, output hash, tolerance envelope, and falsifier. Without those components, OC Core 1.3.2 uses structural or operator-signature language.

Package Safety

The public package excludes private paths, credentials, local machine references, private reviewer-source material, private model transcripts, nonpublic template payloads, and unsupported publication actions. The release ZIP is built from an explicit bundle manifest and a nonpublic-template gate.

Reading Paths

Reviewer	Recommended path
Owner	Owner decision memo, release dossier, no-send manifest, then this companion.
Mathematical reviewer	Master monograph definitions, theorem roadmap, proof machinery, closure ledger, then claim support map.
Domain reviewer	K0–K12 domain projection packet, replay manifests, DRT strict salvage, then methods table.
Reproducibility reviewer	Manifest, checksums, package ZIP, data manifest, simulation results, and PDF quality report.
Publication reviewer	LRGEF state, Parfit report, owner approval packet, release notes, and postflight checklist.

Proven, Replayed, Future

Class	1.3.2 treatment
Proven or evidence-closed	Core definitions, theorem-fate rows, source-audited support classes, and the eighty-two terminal release-blocker rows.
Replayed	K0–K12 projection routes, simulation-result manifests, domain replay status, and DRT workflow/salvage checks where applicable.
Future work	Stronger OC 1.4 formalization, new empirical wins, IP-sensitive race-hardening material, and new domain publication lines.

Evidence Ledger Contract

Each release-visible assertion must name its support surface. If the support is a proof route, the route must bind to a theorem or definition in the master monograph. If the support is replay, the package must expose fixed input, code route, result hash, tolerance, and falsifier. If the support is structural only, the text must say so and must not borrow authority from quantitative prediction language.

Failure Modes That Must Remain Visible

- a theorem label without a proof route becomes a future theorem task, not a release fact;
- a numerical statement without replay support becomes structural context, not a prediction;
- a domain bridge without protocol remains bounded support, not domain replacement;
- a reviewer response without release-visible evidence remains private process, not public science;
- a publication action without owner unlock remains unlawful, even when the science gates pass.

Methods Appendix A: Package Build Contract

Step	Expected result
Build primary PDFs	Six current 1.3.2 PDFs are copied from substantive source PDFs into the artifact directory.
Generate support map	Risky words such as prediction, promoted, held-out, theorem-native, and numerical-table receive release-visible support rows.
Audit packets	All public research packets remain bounded support/frontier material and do not alter canonical claims.
Build manifest	Every ZIP member receives source path, role, size, and SHA256.
Evaluate gates	LRGEF, Parfit, PDF quality, ZIP integrity, and science terminality all pass while no-send remains true.

Methods Appendix B: Reproducibility Reviewer Checklist

1. Compare manifest rows with ZIP entries.
2. Verify checksums for a sample of primary PDFs, data files, and research packet manifests.
3. Extract text from all primary PDFs and scan for stale versions, local paths, unresolved references, and private material.
4. Confirm that legacy 1.3 PDFs are not mixed into the 1.3.2 primary package as active artifacts.
5. Confirm that no publication channel action is encoded as already approved.

Methods Appendix C: Scientific Reviewer Checklist

1. Trace each strong claim to claim ledger, theorem route, replay manifest, or structural-only boundary.
2. Inspect the 82-row closure ledger for row-level evidence and release references.
3. Inspect domain projection rows before accepting domain-facing language.
4. Treat simulations as illustration or replay support only according to their declared class.
5. Reject any statement that tries to turn a frontier packet into canonical authority.

Expected Local Verification

Before owner review, the release must pass:

- release machine package in no-publish mode;
- release machine dry-run evaluation;
- Parfitian Cerberus tests;
- PDF text extraction scans;
- public-safety leak scans;
- version consistency across visible pages, metadata, manifests, and release dossier.